i am looking for the communication module chip of microelectronics which is work at 100M ~200M with AM modulation

If you specifically want a **communication IC from STMicroelectronics** that works with **AM-type modulation (ASK/OOK)** and roughly **100–200 MHz carrier**, there are some important realities:

- ST rarely makes chips exactly for **100–200 MHz VHF band**.
- Most of their RF ICs are designed for **ISM bands** like **315 MHz, 433 MHz, 868 MHz, 915 MHz**.
- However, some ST transceivers can **generate ASK/OOK modulation** and operate in nearby programmable ranges.

Below are the closest practical chips.

---

1. S2-LP RF Transceiver (Recommended ST chip)

This is one of the main RF communication ICs from ST.

Key specs

- Frequency ranges:
  - **413–479 MHz**
  - **452–527 MHz**
  - **826–958 MHz**
- Modulation:
  - **ASK**
  - **OOK**
  - FSK / GFSK
- Data rate:
  - **0.1 kbps – 500 kbps**
- Output power:
  - up to **+16 dBm**

- Sensitivity:
  - about **–130 dBm** STMicroelectron… +1

This chip is commonly used for:

- industrial wireless
- smart meters
- sensors
- remote control

✔ Supports **AM-style modulation (ASK/OOK)**
× But lowest frequency ≈ **413 MHz**, not 100–200 MHz.

---

2. S2-LPTX RF Transmitter

This is a **transmit-only** version of S2-LP.

Features:

- Frequency:
  - **413–479 MHz**
  - **826–958 MHz**
- Modulation:
  - ASK / OOK / FSK
- Data rate:
  - up to **500 kbps** STMicroelectron…

Used for:

- remote control
- wireless sensors
- simple telemetry.

Important: Why you cannot easily find 100–200 MHz chips

The **100–200 MHz band is not commonly used for modern IC RF modules** because:

- many parts of this band are allocated for **VHF radio / TV / aviation / military**
- IC vendors prefer **ISM bands (315 / 433 / 868 / 915 MHz)** for legal reasons.

So most RF IC vendors (ST, TI, Silicon Labs) avoid making chips for that band.

If you truly need 100–200 MHz AM

Usually engineers use one of these approaches:

1. **PLL synthesizer + VCO**
2. **Direct RF oscillator + AM modulator**
3. **Software Defined Radio (SDR) IC**
4. RF IC + **frequency divider/mixer**

✅ **Practical suggestion:**

If you can change frequency, use:

- **433 MHz ASK**
- **315 MHz ASK**

because chips and modules are widely available.

💡 If you want, tell me:

- **transmitter or receiver or both**
- **range (meters / km)**
- **data rate**
- **analog AM or digital ASK**

Then I can suggest **exact chips (ST, TI, Silicon Labs)** that fit your design.

Sources

give me the transreceiver ic from Microelectronics with RDA family for 100 ~ 200 MHz AM modulation

If you want an **RDA Microelectronics RF transceiver IC** that can operate in the **100–200 MHz VHF range**, the main chip in the **RDA family** used for that band is:

1. RDA1846

This is the **most common RDA transceiver used for VHF radios** (handheld radios, walkie-talkies).

Frequency range

- **134 – 174 MHz** (VHF band)
- **200 – 260 MHz**
- **400 – 520 MHz** (UHF band) OpenGD77

So it **covers part of your requested 100–200 MHz range**, especially **134–174 MHz**.

Modulation

- Analog voice modulation (mainly **FM / FSK**)
- Audio baseband modulation through microphone input
- Programmable modulation deviation

Key features

- Fully integrated **RF transceiver**
- Integrated **PLL synthesizer + VCO**
- **RSSI measurement**
- **AGC (automatic gain control)**
- **AFC (automatic frequency control)**
- **DSP audio processing**
- Output power about **7 dBm** typical OpenGD77

Typical applications

- VHF walkie-talkies
- amateur radio handheld devices
- intercom radios
- telemetry radios

Package

- **QFN-32 (5×5 mm)** DatasheetCafe

2. RDA1846S

This is the **newer variant** of the same architecture.

Key differences:

- better DSP
- improved sensitivity
- same **134–174 MHz / 200–260 MHz / 400–520 MHz bands** OpenGD77

Used in many **Baofeng-type radios**.

Important note about AM modulation

The **RDA1846 family does not directly support RF AM modulation**.

It is designed mainly for:

- **FM analog voice**
- **FSK digital data**

The chip performs:

- **audio → FM RF modulation**
- **RF → audio demodulation**

So if you truly need **RF AM transmission**, you typically need:

- external **AM modulator**
- or use **audio AM before FM stage** (not true RF AM)

✅ Conclusion

For **RDA Microelectronics chips covering ~100–200 MHz**, the practical solution is:

IC	Type	Frequency
RDA1846	RF transceiver	134–174 MHz

ChatGPT Free offer ×

💡 If you want, I can also show you **3 more RDA chips used in VHF radios (RDA1845 / RDA1850 / RDA8810)** and explain **which ones can really cover 100–200 MHz**.
Some of them are **much easier to design with than RDA1846.**

Sources

yes please

Here's a list of **RDA Microelectronics transceiver ICs in the RDA family** that *might* be relevant to your need for VHF (100–200 MHz) communication — including a clarification on what each part actually supports:

📡 1. RDA1846 — RF VHF/UHF Transceiver

Type: Single-chip transceiver (TX+RX)

Frequency coverage:

- **134–174 MHz** (VHF)
- **200–260 MHz** (VHF/UHF)
- **400–520 MHz** (UHF) pdf.datasheet.live +1

Features

- Integrated **frequency synthesizer + VCO**
- DSP processing for baseband
- On-chip PA (≈8 dBm) and LNA
- Low external parts count
- 12.5 kHz / 25 kHz channel spacing pdf.datasheet.live

Notes

- Designed for **walkie-talkies / handheld two-way radios**. pdf.datasheet.live
- Supports FM modulation in normal use; **AM modulation isn't directly supported** at RF (needs external stage).

- Programmable LO covers your band *only starting at ~134 MHz*, not down to 100 MHz. PDF Archive

📌 **Best fit in RDA family for your 100–200 MHz R-X/T-X requirement** — if you only need above ~134 MHz and can handle FM/FSK internally. pdf.datasheet.live

---

📡 2. RDA1846S — Improved variant of RDA1846

Type: Single-chip transceiver (TX+RX)

Frequency coverage:

- **134–174 MHz**
- **200–260 MHz**
- **400–520 MHz** opengd77.com

Differences vs RDA1846

- Very similar feature set
- Slight improvements in DSP and sensitivity
- Same band coverage (still not below ~134 MHz) opengd77.com

Use-case:

- Walkie-talkies
- Amateur / public-service radios
- Two-way communication where integration and size matter

Limitations:

- AM at the RF level is *notnatively* supported — this is predominantly for FM/FSK style modulation. opengd77.com

---

× 3. RDA1845 — Similar transceiver but in different bands

The **RDA1845** exists and is a fully-integrated single-chip transceiver, but it is typically noted in documentation for **300–500 MHz** band and *not* documented for 100–200 MHz. DatasheetGO

So it **doesn't help your 100–200 MHz requirement**.

× RDA8810 or similar high-band RDA chips

- **RDA8810** editions you find referenced online are usually **quad-band GSM cellular modem chips**, not standalone RF transceivers for general RF communication. They are *not suitable* for VHF AM/FSK type links. ElectroDragon
- Most RDA SoCs focus on cellular / Bluetooth / UHF ISM bands, not low-VHF analog. DatasheetGO

🔎 Summary — What's in the RDA family for 100-200 MHz

Part Number	TX+RX	Frequency Coverage	Notes
RDA1846	✔	**134–174 MHz**, 200–260 MHz, 400–520 MHz	Best general choice in RDA family for ~VHF; FM-centric
RDA1846S	✔	Same as above	Improved RDA1846 variant
RDA1845	✔	~300–500 MHz	Too high for 100–200 MHz band
RDA8810	✘	GSM / cellular	Not a general RF transceiver

⚠️ Important Modulation Notes

None of these **RDA transceiver ICs natively support pure **AM at RF** like classic analog AM transmitters. They are mainly designed for:

- **FM modulation (narrowband FM)**
- **FSK/GFSK digital** — for voice/data radios.

For true **AM RF modulation**, you typically need either:

1. An external **AM modulator stage** between your RDA's IF and RF chains.
2. A different IC family designed for **ASK/OOK modulation** or wideband VCO/PLL.

⚙️ Recommendation if AM is required

If you *must* do real 100–200 MHz **AM (amplitude modulation)**:

- Use a **wideband RF synthesizer or VCO/PLL chip** (e.g., that can be tuned 50–300 MHz).
- Pair it with a **multiplier / mixer / modulator** to impose AM.
- A popular non-RDA example in that space would be **Silicon Labs wideband transceivers** (e.g. SI4463 family covers 142–1050 MHz) — *not native AM either, but programmable ASK/OOK*.

If you want, I can list **non-RDA RF transceiver ICs that do support ASK/OOK around 100–200 MHz**, including exact parts and datasheets — just let me know!

Sources